Feedback alignment

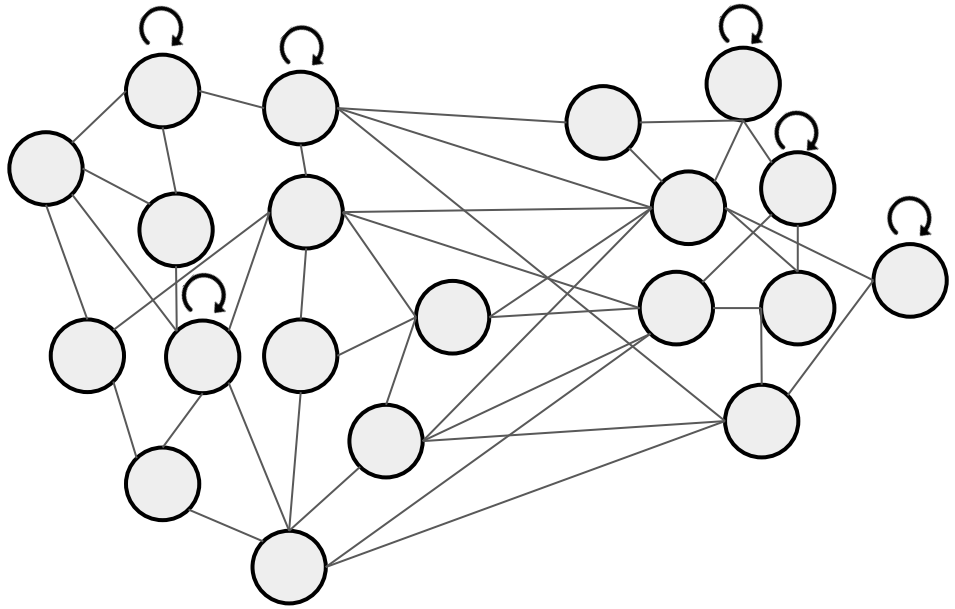
Tutorial 2.2
Learning in deep artificial and biological neuronal networks

Elvis Nava

11.10.2023

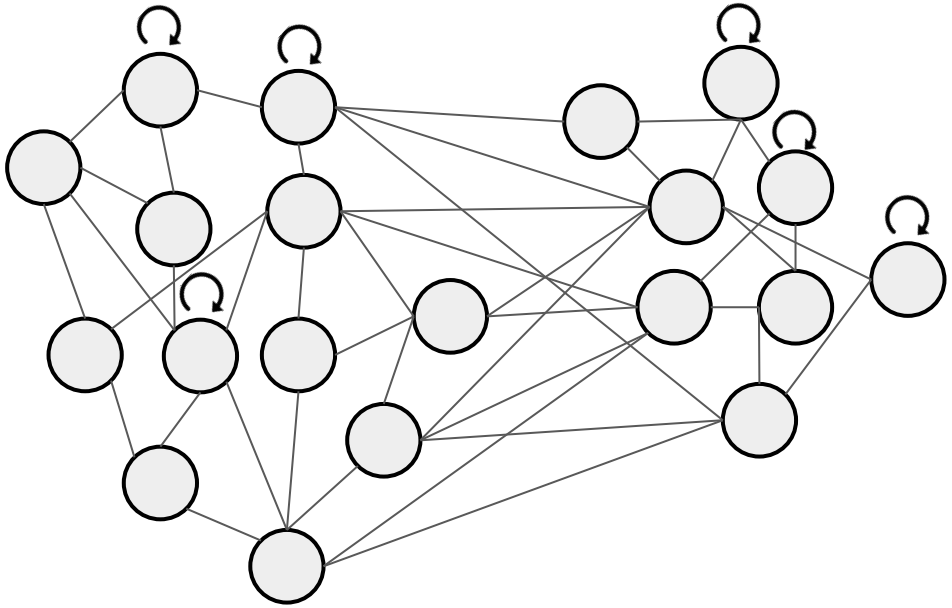
The credit assignment problem



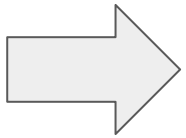


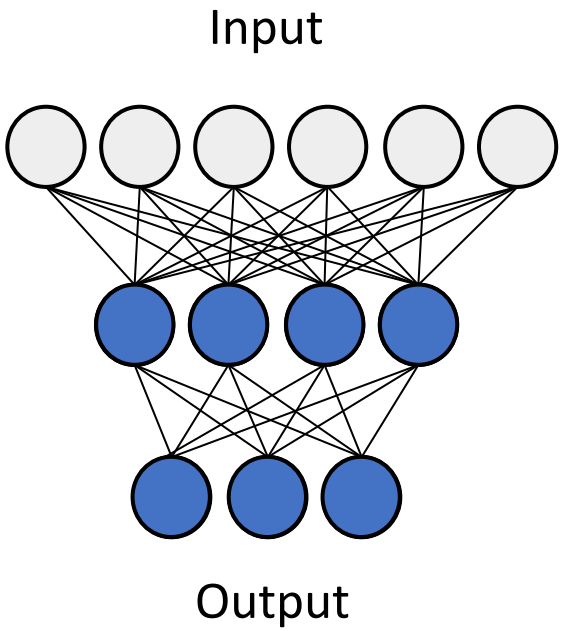
“ How does the strength of a synapse need to be changed to improve the system’s global behaviour?”

Credit assignment in supervised neural networks



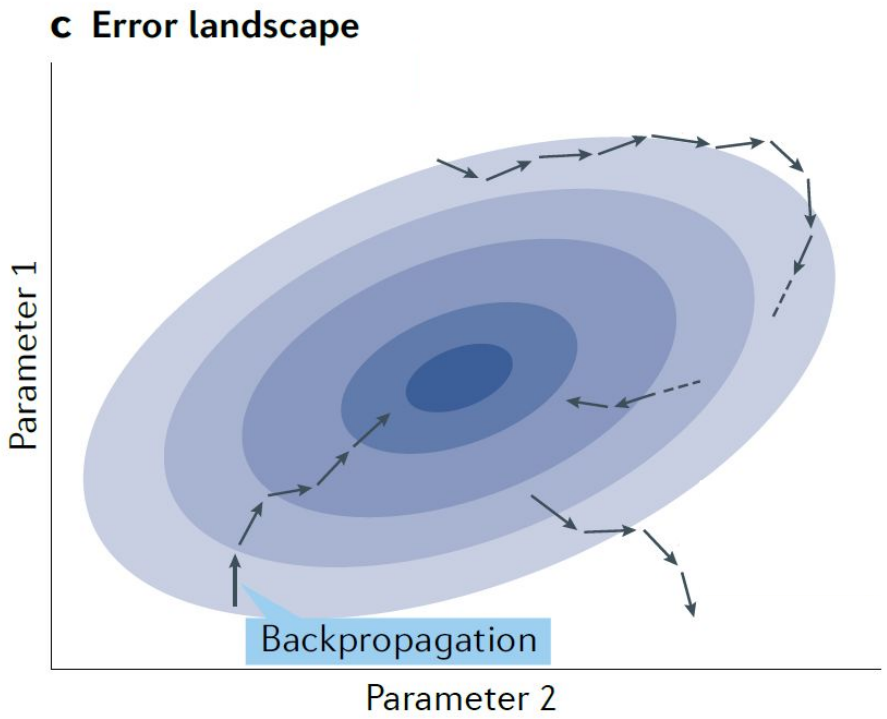
Temporal + spatial
credit assignment

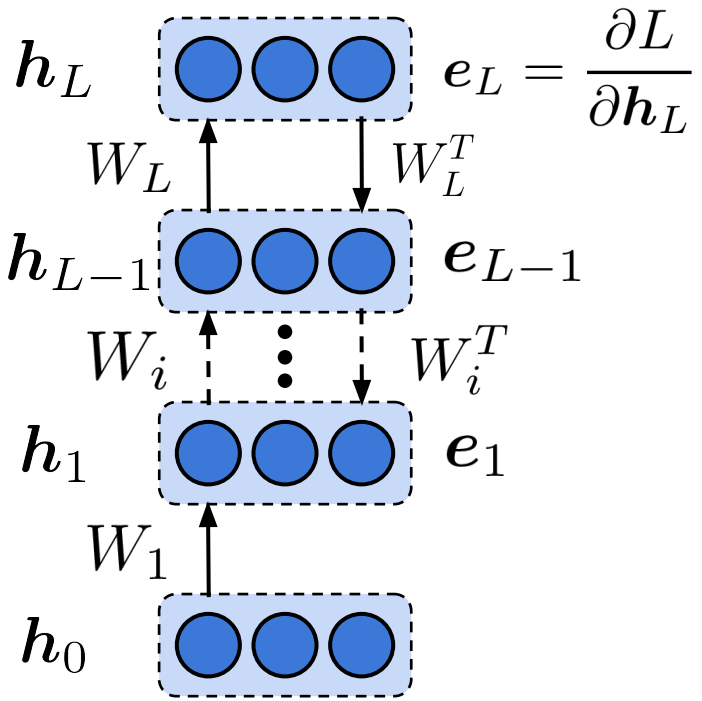




Spatial credit assignment

Error backpropagation = gradient descent





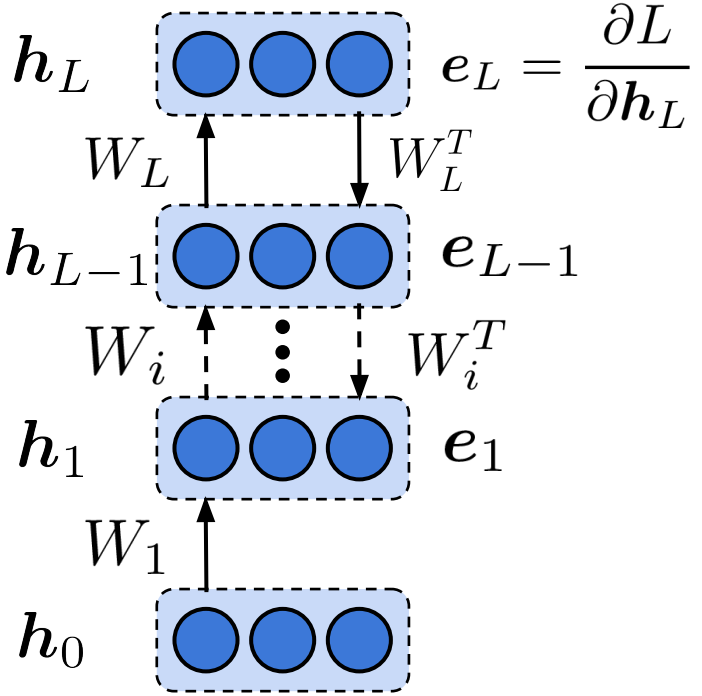
Lillicrap and Hinton 2020

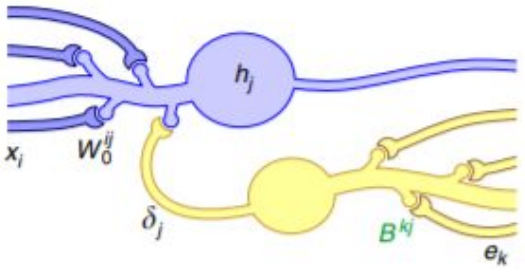
Could our brain do backpropagation?

Main issues:

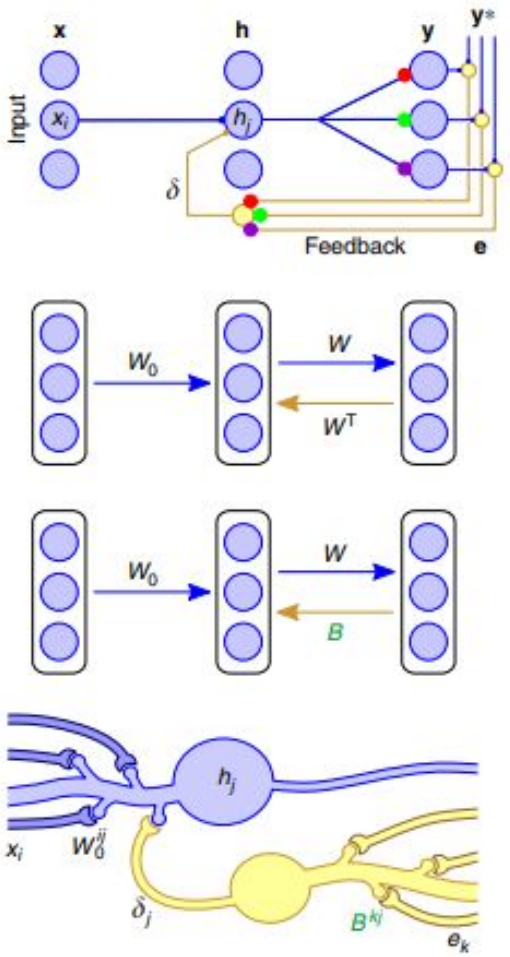
1. Symmetric feedback weights needed
2. Layer-wise feedback path needed
3. Need for the transmission of signed error signals
4. Separate phases needed
5. How is it wired in the brain?

Even for ‘simple’ feedforward neural networks, it is not clear how the brain could do credit assignment in a ‘biologically plausible manner’



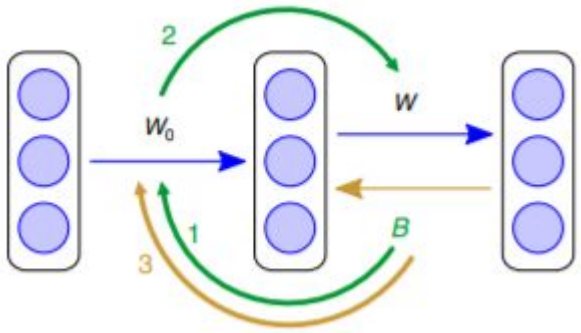


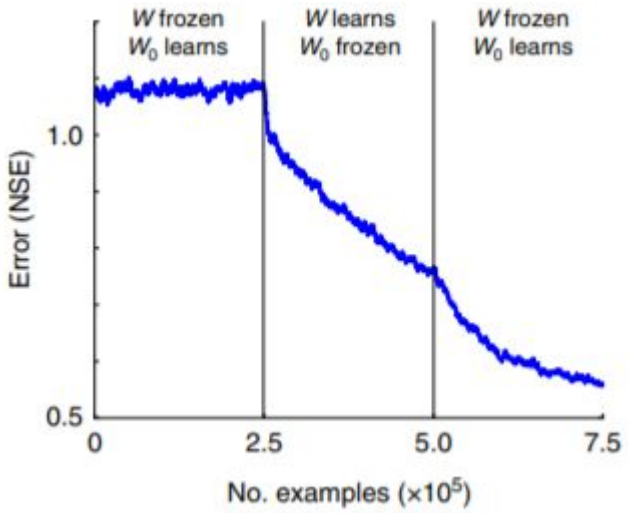
Feedback alignment as solution for the weight transport problem



Lillicrap et al. 'Random synaptic feedback weights support error backpropagation for deep learning' *Nature communications* 2016

Why does it work?





Lillicrap et al. 'Random synaptic feedback weights support error backpropagation for deep learning' *Nature communications* 2016

Implications of FA on the weights

- The forward weights need to fulfill 2 tasks:
  - Reduce the error (like in backpropagation)
  - Align with the random feedback weights

➡ FA can be interpreted as a regularizer

➡ Redundancy needed in the forward weights

Shortcomings of FA

- Scalable?
  - FA works on MNIST, but has trouble with some more complicated datasets such as ImageNet
  - Research ongoing on how to implement FA in more special architectures (e.g. convolutional nets, recurrent nets, …)
- Still not completely biologically plausible
  - Distinct clocked phases
  - Feedback weights are not plastic
  - …

Extensions of FA

- Machine learning side

Deep Learning without Weight Transport

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University of Toronto, Triage

Collin Wilson
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DeepMind

Timothy Lillicrap
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Direct Feedback Alignment Provides Learning in Deep Neural Networks

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- Towards more biological plausibility

Dendritic cortical microcircuits approximate the backpropagation algorithm

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Towards deep learning with segregated dendrites

Jordan Guerguiev[1,2], Timothy P Lillicrap[3], Blake A Richards[1,2,4]*